A Two-Gate Test for Structural Edge in Systematic Trading Strategies

Sofiane Boutgajouft
OrgaX LLC

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Abstract

We propose a two-gate framework for assessing structural edge in systematic trading strategies from realised trade outcomes, without requiring disclosure of the trading signal. Gate 1 screens the mean of a prespecified trade outcome using a t-statistic threshold of three. Gate 2 compares outcome distributions across instruments, volatility regimes and time using normalised pairwise Wasserstein-1 distances and partition-specific Gaussian Monte Carlo reference thresholds. The framework separates evidence of positive average outcomes from the absence of detected distributional differences. Empirical illustrations comprise a historical foreign-exchange application and eight fixed strategies on eight index-tracking ETFs over 2008–2025. The FX strategy is reported to clear both gates. In the ETF panel, RSI(2) clears Gate 1 but exceeds the cross-instrument reference threshold of Gate 2; the remaining seven rules do not reach the mean threshold. These exploratory results illustrate the complementary information supplied by the two gates. The framework supports transparent strategy due diligence, subject to the chosen outcome definitions, reference calibration and dependence between trades. A Gate 2 pass indicates no detected discrepancy under the stated reference, not established invariance or factor-adjusted alpha.

Keywords: structural edge; invariance test; Wasserstein distance; strategy validation; black-box testing; strategy due diligence

JEL codes: C12; C14; C58; G11; G12

1 Introduction

Allocators routinely face a strategy whose track record is observable but whose signal is not. A central question follows: is the strategy's edge structural, invariant across instruments, volatility regimes, and time, or regime-contingent, concentrated in conditions that may not persist and prone to post-publication decay (McLean and Pontiff 2016)? The distinction governs capital allocation, yet it is typically assessed informally by inspecting performance tables broken down by market regime.

This paper formalizes that practice into a Gaussian-reference diagnostic that operates on trade-level output alone, with no access to the signal-generating mechanism. We treat the per-trade R-multiple distribution as the object of study and ask whether it is statistically invariant across three partitions: instrument, volatility regime, and time. The building blocks are established: Wasserstein distances for distributional comparison (Villani 2009; Peyré and Cuturi 2019), and Wasserstein-based goodness-of-fit testing with invariance reduction and calibrated finite-sample critical values (Hallin, Mordant, and Segers 2021).

A distributional score alone cannot establish a positive mean: adding the same constant to

every outcome leaves the normalised Wasserstein score unchanged while changing the mean. Proposition 1 makes this separation explicit. We therefore use two complementary screens. Gate 1 compares a prespecified mean-outcome statistic with a threshold of three, motivated by Harvey, Liu, and Zhu (2016). Gate 2 compares distributional discrepancies across instruments, regimes and time with an explicit Gaussian reference. Clearing both defines an operational classification under the stated conventions; it is not a proof of a structural edge.

Contribution. The statistical objects the test uses are not new. Its contribution is the setting it addresses: third-party due diligence on a strategy whose mechanism is not disclosed, where the only observable object is the realised trade sequence. Other validation tools address complementary questions. Backtest-overfitting diagnostics (Bailey et al. 2017) and multiple-testing corrections (Harvey, Liu, and Zhu 2016) both require the researcher's trial history, which an external evaluator does not have and cannot recover from a single realised sequence. Factor attribution requires a return series aligned to factor dates and answers a different question: whether the return is compensation for known exposures, not whether the edge is invariant across the conditions in which it was earned. We assemble established components into a documented screening protocol for the signal-blind setting. Appendix C gives the score's affine-invariance and payoff-comparison properties, a population-limit result under an idealised sampling model, and the assumptions underlying its Gaussian location-scale reference. These mathematical results clarify the same implemented protocol.

Reproducibility. The empirical section uses free public sources. The publicly specified strategies are reconstructed on daily OANDA price series, with policy rates from FRED and the volatility-regime partition from the CBOE VIX. The tested strategy has been regenerated on the same price series. The public repository identifies the reference implementation; recovered trade exports and their verification reports accompany this revision. Its signal remains undisclosed, which is the setting the test addresses.

2 The Two-Gate Test

Inputs and scope

The trade sequence. A strategy produces a finite sequence of trades indexed i = 1, . . . , N. Trade i has a realised profit-and-loss PnL_i and a risk amount Risk_i ¿ 0 committed at entry, before the outcome is known. The R-multiple of trade i is the ratio

$$R_i = \frac{\mathrm{PnL}_i}{\mathrm{Risk}_i}. \tag{1}$$

Thus $R_i = -1$ means a loss equal to the chosen risk unit, and $R_i = 3$ means a gain of three units. The risk unit is fixed before the outcome is observed; it need not be a guaranteed maximum loss. This normalisation defines the outcome being compared. It does not make different risk conventions economically equivalent: trade-specific denominators can change both the mean and the distributional score. Write

$$\bar{R} = N^{-1}\sum_{i=1}^{N} R_i, \qquad \hat{\sigma}_R^2 = (N-1)^{-1}\sum_{i=1}^{N}(R_i - \bar{R})^2.$$

The exact score below requires finite observations, $N \geq 2$, and $\hat{\sigma}_R > 0$.

Partition labels. For each dimension $d \in \mathcal{D} = \{P1, P2, P3\}$, let $L_{d,i} \in \{1, \ldots, K_d\}$ label trade i. The expected group set is specified in the protocol. Define

$$G_{d,j} = \{i : L_{d,i} = j\}, \qquad n_{d,j} = |G_{d,j}|, \qquad \widehat{F}_{d,j}(x) = \frac{1}{n_{d,j}} \sum_{i \in G_{d,j}} \mathbf{1}\{R_i \le x\}. \tag{2}$$

P1 groups trades by instrument, P2 by volatility regime, and P3 by entry-time block. The regime definition, information timestamp, block boundaries and expected instruments must be recorded before interpreting an assessment. The historical FX illustration uses a retrospective full-sample median split of VIX; such a threshold is not known at each historical entry date. The ETF application instead uses the preceding US equity session's VIX with a fixed boundary of 20 and calendar blocks 2008–2013, 2014–2019 and 2020–2025. A retrospective partition is a descriptive sample analysis, not a prospective trading input.

An exact score is undefined if $K_d < 2$, an expected group is empty, or the pooled standard deviation is zero. In the ETF implementation, every expected group must additionally contain at least twenty trades. This guard is an implementation requirement, not a theorem about power. An unavailable or undersized group produces an *indeterminate* distributional assessment; it must not be replaced by a score of zero or silently removed.

What the tester does not observe. The test consumes $\{R_i\}$ and the three label vectors, and nothing else. The signal that generated the trades, the entry and exit rules, the position sizing, and the parameters of the strategy are all unobserved. This restriction is what allows a third party to validate a strategy whose mechanism the owner will not disclose.

Scope. The test applies to strategies defined by a signal producing discrete trades. A static position held throughout the sample has no entry or exit signal and therefore no trade structure to partition; such positions lie outside the scope of the test, and their return reflects a directional risk premium rather than a systematic edge.

Gate 1 screens the mean of a prespecified outcome

Let X_i be the outcome chosen for Gate 1, with a finite, nonzero sample standard deviation. Define

$$T_N(X) = \frac{\sqrt{N}\,\bar{X}}{s_X}, \qquad \bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i, \qquad s_X^2 = \frac{1}{N-1}\sum_{i=1}^{N}(X_i - \bar{X})^2. \tag{3}$$

The mean hypothesis of interest is $H_0^E : \mu_X \le 0$ against $H_1^E : \mu_X > 0$. The reported screen is $T_N(X) \ge t^\star$, with $t^\star = 3$. This fixed bar is motivated by Harvey, Liu, and Zhu (2016); it is not a universal significance level or a correction for an unknown history of strategy selection. Under independent and identically distributed Gaussian outcomes with common mean μ_X and variance $\sigma_X^2 > 0$, the boundary case $\mu_X = 0$ yields a Student t_{N-1} law. That exact statement does not apply automatically to overlapping financial trades.

The choice of X_i is part of the protocol and must not be switched after observing which definition clears the threshold. In the ETF experiments and the proprietary FX illustration, $X_i = R_i$. The historical public FX benchmarks instead use signed raw instrument returns over a fixed twenty-session window. Those recorded conventions are reported explicitly; their t-statistics are not a controlled comparison of identical outcome definitions. In particular, a t-statistic on unscaled net returns is not a replacement for the prespecified t-statistic on R after results are inspected.

A common positive rescaling of every outcome leaves T_N unchanged, whereas trade-specific risk rescaling generally does not. For any common $a \neq 0$ and $b \in \mathbb{R}$,

$$T_N(aX + b) = \operatorname{sgn}(a) T_N(X) + \frac{b\sqrt{N}}{|a| s_X}. \tag{4}$$

This contrasts with the common-affine invariance of Gate 2 and explains why a distributional score alone cannot assess a positive mean.

Gate 2 tests invariance

Gate 2 measures discrepancies between conditional outcome distributions. For a population interpretation, write $F_{d,j} = \mathcal{L}(R \mid L_d = j)$, assuming these conditional laws exist and have finite first moments. The equality hypothesis is $H_{0,d} : F_{d,1} = \cdots = F_{d,K_d}$. A difference may concern the mean, dispersion or tails; an elevated score need not imply that the mean edge itself depends on that partition.

Comparing two groups. Let $\hat{F}_{d,j}$ denote the empirical distribution of the R-multiples of trades in group j of dimension d. For two univariate distributions μ and ν, the Wasserstein-1 distance is

$$W_1(\mu, \nu) = \int_0^1 \left| F_\mu^{-1}(u) - F_\nu^{-1}(u) \right| \, du. \tag{5}$$

This is the mean absolute gap between generalised quantile functions (Villani 2009; Peyré and Cuturi 2019). For empirical measures, let $z_1 < \cdots < z_M$ be the sorted distinct values in the two groups' combined support. An equivalent exact expression is

$$W_1(\widehat{F}_{d,j}, \widehat{F}_{d,k}) = \sum_{\ell=1}^{M-1} \left| \widehat{F}_{d,j}(z_\ell) - \widehat{F}_{d,k}(z_\ell) \right| (z_{\ell+1} - z_\ell). \tag{6}$$

The sum is zero when $M = 1$. This formula handles unequal group sizes and ties and is equivalent to exact integration of empirical step quantiles. It describes the exact ETF statistic. The historical FX implementation uses a finite interpolated quantile grid and is a numerical approximation, so exact zero-characterisations should not be transferred to that approximation without checking it.

Aggregating across groups. A dimension with K_d groups admits $K_d(K_d - 1)/2$ pairwise comparisons. The structural invariance score of dimension d is their average, divided by the dispersion of the whole sequence:

$$\mathrm{SIS}_{d,N} = \frac{1}{\hat{\sigma}_R} \cdot \frac{2}{K_d(K_d - 1)} \sum_{j<k} W_1\left(\hat{F}_{d,j}, \hat{F}_{d,k}\right). \tag{7}$$

When N is understood, write $\mathrm{SIS}_d = \mathrm{SIS}_{d,N}$. The score is an equally weighted average over group pairs, expressed in units of pooled outcome dispersion. Proposition 1 establishes nonnegativity, the exact zero case and invariance under a *common* nondegenerate affine transformation. Proposition 2 relates it to differences in group means and expected downside magnitudes. Neither result makes it invariant to arbitrary trade-specific risk conventions.

Worked example. The volatility dimension P2 has $K_{P2} = 2$ groups and therefore one pairwise distance. The historical FX implementation reports $\hat{\sigma}_R = 1.271$ and $\mathrm{SIS}_{P2} = 0.059$ for the tested

strategy, 0.001 for the 5-day mean-reversion benchmark, and 0.135 for Bollinger reversal. These are previously reported values from the interpolated quantile-grid implementation, rather than newly verified exact empirical distances. Judging their magnitude also requires their corresponding reference thresholds.

Gaussian reference and finite simulation

Fix the observed labels $\lambda_d = (L_{d,1}, \ldots, L_{d,N})$ and group sizes. For each $b = 1, \ldots, B$, independently generate $Z^{(b)} \sim \mathcal{N}_N(0, I_N)$ and compute

$$S_d^{(b)} = \mathrm{SIS}_{d,N}(Z^{(b)}; \lambda_d), \qquad \widehat{\tau}_{d,N} = \widehat{Q}_{0.95}^{\mathrm{linear}}\big(S_d^{(1)}, \ldots, S_d^{(B)}\big), \qquad B = 10{,}000. \tag{8}$$

Each replicate uses its own pooled sample standard deviation. The ETF tables use the linear-interpolated empirical quantile implemented in the notebooks. For a population reference, define

$$\tau_{d,N}^{G}(1-\alpha) = \inf\{u : \Pr_{G}(\mathrm{SIS}_{d,N}(Z; \lambda_d) \le u) \ge 1 - \alpha\}.$$

The simulated threshold estimates this reference quantile; its interpolation convention is not an exact finite-simulation test.

The group-size vector $(n_{d,1}, \ldots, n_{d,K_d})$ determines the reference distribution. Its dependence on sample size and partition structure does not imply universal monotonicity in N or K_d. Proposition 3 states the restricted conditional Gaussian model under which location and scale are nuisance parameters removed by the score. Equality of arbitrary non-Gaussian group distributions, or equality with serially dependent observations, does not imply that model. Gaussian reference thresholds are therefore not distribution-free and do not provide a general type-I error guarantee for trading outcomes.

The reported Gate 2 convention is

$$\mathrm{SIS}_{d,N} < \widehat{\tau}_{d,N} \quad \text{for every evaluable } d \in \mathcal{D}, \quad \text{with all expected dimensions evaluable.} \tag{9}$$

A component below its threshold means that the chosen reference discrepancy was not exceeded. It is not a confidence statement that two population distributions are equal or economically equivalent. The three component thresholds are retained exactly as implemented. Their conjunction is an operational classification; no overall five-percent error guarantee or equivalence claim is attached to it.

Theorem 1 gives consistency against fixed distributional alternatives in an idealised independent sampling model with positive limiting group shares. It does not supply a finite-sample power guarantee, a universal minimum trade count, or validity under dependence.

Classification

For reporting purposes, a strategy clears the operational two-gate rule if and only if:

$$T_N(X) \ge 3 \quad \text{and} \quad \mathrm{SIS}_{d,N} < \widehat{\tau}_{d,N} \quad \forall d \in \mathcal{D}. \tag{10}$$

The combined status is indeterminate if either required statistic is unavailable. Otherwise the rule is cleared only when both conditions hold. By common-translation invariance, the same Gate

2 score is compatible with positive, zero or negative mean outcomes. Conversely, a high pooled mean can coexist with large between-group discrepancies. These observations establish the logical distinction between the screens, not a proof that their intersection identifies a causal or permanent source of return.

Table 1 Interpretation of failing components

Failure	Reading
$T_N(X)$ ¡ 3	Gate 1 not cleared: the prescribed evidence threshold is not reached
$\mathrm{SIS}_{P2} \geq \widehat{\tau}_{P2}$	Between-regime distributional discrepancy reaches or exceeds its Gaussian reference
$\mathrm{SIS}_{P3} \geq \widehat{\tau}_{P3}$	Between-period distributional discrepancy reaches or exceeds its Gaussian reference
$\mathrm{SIS}_{P1} \geq \widehat{\tau}_{P1}$	Between-instrument distributional discrepancy reaches or exceeds its Gaussian reference

Gate 1 determines eligibility for the combined rule. Gate 2 may still be reported descriptively when Gate 1 is not cleared, because distributional heterogeneity remains informative independently of the pooled mean.

Relation to existing validation tools

The test addresses a setting the standard toolkit does not cover: an evaluator holds a strategy's realised trade sequence and nothing else.

Backtest-overfitting diagnostics. Bailey et al. (2017) estimate the probability that an apparent edge arises from in-sample selection. Such diagnostics require the researcher's trial history, the set of configurations examined, which an external evaluator does not have and cannot recover from a single realised sequence. The two-gate test is silent on selection bias and should be read alongside such diagnostics.

Multiple-testing corrections. Harvey, Liu, and Zhu (2016) likewise require the number of trials to deflate the reported statistic. We borrow their t $\geq$ 3 bar for Gate 1 but cannot apply the deflation itself, for the same reason.

Factor attribution. This regresses a return series on known factors and asks how much of the return they explain. It requires a return series aligned to factor dates rather than a trade sequence, and asks whether the return is compensation for known exposures, not whether the edge is invariant across the conditions in which it was earned. The two are complementary. The two-gate test characterises robustness and does not decompose an edge into alpha and beta; a strategy with a positive mean and sufficiently small distributional discrepancies may clear both gates while carrying stable factor exposure. Establishing alpha in the asset-pricing sense requires a separate factor attribution.

Interpretation and assumptions

The reported Gate 1 statistic uses the conventional standard error $s_X/\sqrt{N}$. For any square-integrable outcome vector, however,

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2}\left[\sum_{i=1}^{N}\mathrm{Var}(X_i) + 2\sum_{i<j}\mathrm{Cov}(X_i, X_j)\right]. \tag{11}$$

This is the variance-of-a-sum identity; independence removes the covariance terms. Simultaneous ETF exposures and overlapping holding windows can produce nonzero terms, so a large trade count does not establish a valid independent-observation standard error. Dependence-aware inference would require an explicitly justified calendar-cluster or time-series variance estimator and its own assumptions. No such correction is included in the reported t-statistics, and reordering same-date trades does not solve the problem.

Gate 2 compares observed distances with Gaussian reference thresholds. A component below its threshold is an absence of detected deviation under this convention, not an equivalence test. No joint significance level is claimed for passing all three components. The choices of partition, payoff definition and reference distribution remain part of the protocol and require sensitivity analysis.

The FX reference application and the additional ETF experiment use different outcome constructions. The original FX implementation approximates the quantile integral on a finite grid, whereas the additional ETF notebooks compute empirical Wasserstein distances exactly. Their numerical reference values should therefore not be pooled or treated as a single identically calibrated experiment.

3 Data and Strategies

Data

The historical application uses daily price data. Spot foreign-exchange series for seven major pairs (EURUSD, GBPUSD, AUDUSD, NZDUSD, USDJPY, USDCHF, EURGBP) are taken from OANDA via QuantConnect. Short-term policy rates used to construct the carry benchmark are taken from FRED. The historical volatility-regime partition P2 uses the CBOE VIX and a retrospective in-sample median of 17.49. The sample runs from January 2008 to December 2025; the first trade of the tested strategy occurs in February 2009, after its warm-up requirement is met.

The experiments use publicly obtainable price and rate sources. The availability and licensing of the source datasets remain separate from the supplied derived trade-level outputs.

The tested strategy

The strategy under test is a proprietary systematic foreign-exchange strategy trading the seven pairs above. It identifies structural equilibrium levels by swing-point analysis and enters on displacement events around them; its parameters are proprietary and its signal is undisclosed. Only its realised R-multiple output enters the test.

Its output has been regenerated on the OANDA price series used for the benchmarks, replacing the commercial-terminal series of the previous version. The regeneration also corrected a look-ahead bias in the implementation: stop and target resolution had been permitted on the same bar as entry and is now deferred to the following bar. The corrected sequence contains N = 582 trades, with mean R-multiple $\bar{R} = 0.192$ and $\hat{\sigma}_R = 1.271$. The historical FX results in Table 2 retain the values reported in that version.

Benchmark strategies

To assess discriminant validity, we reconstruct publicly specified trading strategies on the same instruments and sample. Each is defined by a transparent rule, replicable from published descriptions, and uses its conventional published parameter setting; none is tuned on the sample.

Mean reversion. 5-day reversal (fade the sign of the trailing 5-day return); Bollinger reversal (fade a two-standard-deviation deviation from the 20-day mean); RSI(2) reversal.

Trend. 20/60 and 10/50 moving-average crossovers; MACD (sign of the 12/26 exponential moving-average gap).

Momentum. 60-day time-series momentum; 12-month time-series momentum following Moskowitz, Ooi, and Pedersen (2012); cross-sectional momentum (long the two strongest pairs, short the two weakest, on the trailing 60-day return).

Breakout. Donchian range breaks at 20 and 55 days.

Carry. Cross-sectional carry: long the two highest-yielding currencies and short the two lowest, rebalanced monthly, using published central-bank policy rates.

Reversal cross-sectional. The mirror of cross-sectional momentum on the trailing 5-day return.

Mapping signals to trades. Within each instrument, each signal is mapped to a sequence of non-overlapping trades. Entries sit on a fixed grid of twenty trading days, so the sampling rule does not depend on how quickly a rule exits. The construction prevents overlapping holding windows within an instrument; trades across instruments may still overlap in calendar time. For the invariance components, each entry is assigned a fixed reward-to-risk payoff: the risk unit is $R = c\hat{\sigma}(20)$, with $c = 0.5$ and $\hat{\sigma}(20)$ the trailing 20-day return volatility; a path reaching $+3R$ books $+3$, one reaching $-1R$ books -1, and a 20-day time-stop books 0. Holding this payoff constant across benchmarks keeps the discretised shape of the R-multiple distribution comparable.

This payoff is not used to judge edge. The edge verdict under a fixed payoff depends on the arbitrary stop distance, so benchmark edge is measured on the raw directional return over the same twenty-day window. The tested strategy reaches its payoff through its own exit logic, so its realised R-multiples enter both gates.

A strategy excluded from the panel. An earlier version included a permanently long equity and gold position among the benchmarks. A static buy-and-hold position has no entry or exit signal and therefore no trade structure to partition; it lies outside the stated scope. We discuss the case below because it bears on what the test does and does not establish, but it is not part of the benchmark panel.

4 Results

The empirical evidence contains two separate applications. Table 2 preserves the historical FX results, and Table 3 adds the index-ETF experiment. The latter includes a public rule that clears Gate 1 and then exceeds a component of Gate 2. The outcome conventions and replication status of the two panels are recorded separately; Appendix B gives implementation details for the new experiments.

Table 2 reports the two-gate classification of the historical FX benchmarks and the tested

strategy. Benchmark edge is measured on the pooled raw directional return; the tested strategy's edge is measured on its realised R-multiples. Invariance components use fixed reward-to-risk R-multiples for benchmarks and realised R-multiples for the tested strategy, with per-dimension null thresholds from B = 10,000 Monte-Carlo replications.

Table 2 Two-gate classification over 2008-2025

Strategy	N	t_{edge}	SIS_{P1}	SIS_{P2}	SIS_{P3}	Verdict
FX strategy (tested)	582	3.64	0.136	0.059	0.059	passes rule
Bollinger reversal	1195	2.95	0.086	0.135	0.060	not cleared
60-day momentum	2529	2.24	0.054	0.121	0.076	not cleared
XS momentum (60d)	1583	0.71	0.089	0.078	0.033	not cleared
XS reversal (5d)	1846	0.57	0.059	0.022	0.059	not cleared
20/60 trend	2529	0.53	0.044	0.045	0.067	not cleared
MACD	2529	0.24	0.044	0.013	0.042	not cleared
TS momentum (12m)	1967	0.02	0.057	0.111	0.066	not cleared
Donchian 55	1461	−0.23	0.111	0.022	0.093	not cleared
Carry (policy rates)	1153	−0.33	0.152	0.085	0.050	not cleared
10/50 trend	2529	−0.33	0.052	0.018	0.014	not cleared
RSI(2) reversal	1738	−0.56	0.070	0.037	0.043	not cleared
5-day mean reversion	2528	−0.67	0.046	0.001	0.030	not cleared
Donchian 20	2039	−1.01	0.091	0.021	0.029	not cleared

Notes: Edge for benchmarks is the t-statistic of pooled raw directional returns; for the tested strategy, it is the t-statistic of realised R-multiples. P1: cross-section; P2: volatility regime; P3: time.

Gate 1 separates the tested strategy from the public panel. Within the historical FX benchmark panel of Table 2, no rule attains the prescribed Gate 1 threshold on raw directional returns: t_{edge} ranges from −1.01 to +2.95, all below 3. The panel spans the standard families of mean reversion, trend, momentum, breakout, and carry at conventional parameter settings. The tested strategy attains $t_{\text{edge}} = 3.64$ on the same price series.

Outcome conventions. The historical public benchmarks use raw directional returns for Gate 1 and fixed-payoff R-multiples for Gate 2, whereas the proprietary illustration uses its realised R-multiples in both gates. This follows the recorded protocol. It also limits cross-row comparisons: common scale invariance does not imply that trade-specific payoff transformations preserve a t-statistic. The additional ETF experiment uses net R-multiples in both gates for every rule.

Invariance alone is insufficient. Several benchmarks have low distributional components despite not clearing Gate 1: the MACD rule has $\text{SIS}_{P2} = 0.013$ and the 5-day reversal $\text{SIS}_{P2} = 0.001$, both far below their thresholds. A low distributional discrepancy alone therefore does not establish a positive mean. This is the empirical motivation for keeping the mean-return gate separate.

Gate 2 discriminates among the failures. Three benchmarks show an elevated regime component relative to their own thresholds: Bollinger reversal, 60-day momentum, and 12-month time-series momentum, indicating volatility-regime dependence. These rules do not clear Gate 1, so the Gate 2 results remain descriptive: they identify distributional differences relative to the same reference convention.

The tested strategy clears both gates. With $t_{\text{edge}} = 3.64$ and all three invariance components

below their $P_{95}(H_0)$ thresholds (P1: 0.136 ¡ 0.240; P2: 0.059 ¡ 0.177; P3: 0.059 ¡ 0.184), the tested strategy clears the stated reference rule. This is an operational classification and does not establish distributional equivalence, causal structure, or alpha. A previously reported sensitivity check removes the ten largest R-multiples and gives $t_{\text{edge}} = 3.06$. Its mean R-multiple is positive in each of the three time blocks (+0.130, +0.201, +0.239), though it rises across them, a gradient the temporal component does not find sufficient to reject invariance.

What a passing verdict does and does not establish. A permanently long equity and gold position clears both gates on this sample: its reported mean and distributional components clear the recorded reference thresholds. It is nonetheless an exposure to a directional risk premium rather than an alpha-generating strategy. The two gates separately assess a mean-return screen and distributional stability relative to the reference; they do not decompose it into compensation for known factor exposures and residual return. Establishing alpha in the asset-pricing sense requires a factor attribution, complementary to and outside the scope of a signal-blind test.

Sample size. The tested strategy has N = 582. Appendix A studies regime and instrument partitions; it does not supply a temporal-component power curve or justify a universal N = 600 cutoff. At N = 582, the regime component detects a ten-percentage-point win-rate gap with probability approximately 0.64 by interpolation of Table 4; the corresponding figure for the cross-sectional component is lower. Passing components should therefore be read as an absence of detected deviation rather than as established invariance.

Historical evidence and replication status. Table 2 and Appendix A preserve the numerical results of the original manuscript. They are reported separately from the newly recovered ETF trade sequences and have not been independently replayed as part of this revision. In particular, the historical FX volatility timestamp convention and supplementary sensitivity calculations require their corresponding archived inputs for exact reproduction. This distinction separates the reported reference application from the output-level checks completed for the new experiments.

4.1 Additional public ETF experiments

The ETF results supplement the FX illustration within the same Wasserstein-1/SIS framework. The recorded calculations, thresholds and outcome conventions of each application are retained.

Eight fixed rules were evaluated on DIA, MDY, EWG, EWJ, EWU, EWC, EWA and EWH over 2008–2025. The rules cover short-term mean reversion, trend, breakout and monthly momentum. Signals use information available at the close, with execution at the next opening and explicit costs. All eight ETF rules use net R-multiples for both gates. Their risk convention, portfolio constraints and partitions are described in Appendix B; neither the threshold of three nor the Gaussian-reference rule was adjusted to obtain a pass.

Table 3 provides the additional results. RSI(2) supplies the main contrasting case: with N = 1,023 and t(R) = 3.054 it clears Gate 1, but its instrument component is 0.200093 against a reference threshold of 0.194403. Its regime and time components remain below their respective thresholds. A positive pooled mean screen therefore does not imply that every distributional component clears its reference. The net-return t-statistic of 2.948 is retained as a diagnostic and is not substituted for the prespecified t-statistic on R.

IBS, Bollinger and MACD do not clear Gate 1. Their recovered trade files contain 2,706, 756 and 1,466 trades respectively, with the reported means, t-statistics, SIS values and Gaussian

thresholds reproduced from those files. The three slower rules have insufficient trades in at least one expected instrument group for a full Gate 2 assessment. Their appropriate status is insufficient group data, rather than an observed rejection in that component.

Table 3 Additional index-ETF results

Rule	N	Mean R	t on R	Distributional assessment
RSI(2)	1023	0.079	3.054	P1 exceeded
Donchian20/10	800	0.053	1.035	P2 exceeded
IBS pullback	2706	0.017	1.243	P2, P3 exceeded
Bollinger20/2	756	0.129	2.570	P3 exceeded
MACD12/26/9	1466	0.052	1.522	P2, P3 exceeded
SMA50/200	105	1.466	2.032	P1 insufficient
Momentum252 monthly	103	1.739	2.486	P1 insufficient
Rotation12–1 top3	135	0.615	1.722	P1 insufficient

Notes: Gate 1 requires t(R) $\geq$ 3. P1, P2 and P3 compare instruments, volatility regimes and time blocks. Every expected ETF group must contain at least twenty trades for a complete Gate 2 assessment; this is an implementation guard, not a power guarantee. All eight rule sets are retained. Numerical outcomes are reported under their recorded protocol.

Table 5 in Appendix B gives the three distributional scores and their corresponding reference thresholds for all eight rules.

The eight-rule experiment is exploratory and was assembled after earlier results were inspected. All eight rules are reported, including those with small group sizes. The results do not represent an untouched out-of-sample test or establish that any strategy is universally profitable or unprofitable. The recovered CSVs verify supplied outcomes and calculations, not the source market data or fills. RSI and Donchian are reported from their recorded notebook summaries; their raw trade exports are not included in this revision.

5 Conclusion

We have proposed a Gaussian-reference diagnostic of structural edge that operates on a strategy's trade-level output alone, with no access to the signal. The test combines two complementary gates: significance of the mean return and distributional invariance across instrument, regime, and time. Invariance without an edge is not evidence of structure. In the historical FX panel, benchmark edge is evaluated on raw directional returns; the additional ETF panel uses net R-multiples in both gates.

In the historical FX benchmark panel, no publicly specified rule clears Gate 1 under the recorded outcome conventions, while the proprietary FX illustration is reported to clear both reference screens. In the separate ETF experiment, RSI(2) clears Gate 1 but exceeds the instrument reference threshold of Gate 2. The remaining seven ETF rules do not reach the prescribed t(R) threshold. These findings describe the stated samples and conventions; they do not establish that public strategies lack an edge or that a passing strategy has invariant performance. The reported Monte Carlo study illustrates sample-size sensitivity under a particular synthetic design.

The test complements existing validation tools. It speaks to regime robustness rather than in-sample selection bias and should be read alongside backtest-overfitting diagnostics. It screens

distributional stability rather than decomposing return into known factor exposures and should be read alongside factor attribution. Its practical use is third-party due diligence on strategies whose mechanism is undisclosed. Extending the framework to a broader cross-section of audited live strategies, and exhibiting a live regime-contingent rejection at adequate sample size, are the next steps.

Author declarations

Interest. The author is the founder of OrgaX LLC, which develops the proprietary FX strategy used as the historical illustration. This constitutes a financial interest in that application. **Funding.** No funding was received. **AI assistance.** AI assistance was used in preparing this manuscript.

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Data and code availability

The reference repository is https://github.com/SofianeBoutgajouft/two-gate-test. The accompanying supplement contains research notebooks, recorded experiment summaries, the tested strategy's 582-trade R-multiple sequence, and recovered IBS, Bollinger and MACD exports with verification reports. The signal-generating code remains undisclosed: the test consumes trade outcomes. Source datasets are OANDA prices via QuantConnect, FRED policy rates and CBOE VIX; their licensing and access are separate from these derived outputs. Appendix B identifies the remaining unrecovered exports and the scope of numerical reproduction.

Appendix A. Sample-size sensitivity of Gate 2

The following experiment examines sample-size sensitivity under one specific discrete-payoff alternative; its results are not a universal sample-size recommendation. We simulate sequences with a controlled departure from invariance, apply the gate, and record the rejection rate. The design follows the graded experiment: R-multiples take the fixed-payoff values $\{+3, 0, -1\}$, and the departure is a win-rate gap between groups. The experiment is entirely synthetic, so the calibration does not depend on a particular sample.

Three readings follow. First, the reported zero-gap rejection frequencies range from 0.02 to 0.07 for the regime component and from 0.00 to 0.01 for the cross-sectional component. Thus they do not all lie at or below 0.05. These simulations describe the stated discrete-payoff experiment and do not establish general calibration across outcome distributions. Second, the regime component reaches conventional power against a ten-point win-rate gap between N = 800 and N = 1000; at N = 600 its power is 0.65. The recommendation $N \geq 600$ should therefore be read as the point at which the component detects a ten-point gap roughly two times in three, with $N \approx 1000$ required for the conventional 0.80.

Table 4 Rejection rates by sample size and injected win-rate gap

Panel A P2 with two groups for volatility regime

Gap (pp)	N = 200	300	400	500	600	800	1000	1200
0	.02	.04	.04	.04	.06	.03	.07	.04
5	.10	.13	.14	.17	.18	.25	.29	.35
10	.26	.37	.43	.56	.65	.75	.81	.89
15	.54	.68	.84	.87	.94	.98	.99	1.00
20	.74	.92	.97	.99	1.00	1.00	1.00	1.00

Panel B P1 with seven groups for instrument

Gap (pp)	N = 200	300	400	500	600	800	1000	1200
0	.00	.01	.01	.01	.01	.00	.00	.01
5	.01	.01	.01	.01	.01	.01	.01	.01
10	.03	.03	.04	.03	.04	.05	.08	.07
15	.06	.05	.07	.11	.15	.19	.29	.39
20	.09	.13	.23	.27	.32	.44	.64	.76

Notes: The zero-gap rows report rejection frequencies under this discrete-payoff design. They are not guaranteed to be at or below 0.05 under a Gaussian reference. Values and simulation design are those reported in the original manuscript.

Third, power is low for the seven-group instrument partition in this particular simulation: it reaches 0.04 against a ten-point gap at N = 600, and a twenty-point gap at N = 1200 yields 0.76. Group shares, the placement of the alternative and the averaging of pairwise distances all affect this result. The table does not establish universal monotonicity of the Gaussian threshold or power in the number of groups. A component below its reference at moderate sample size remains an absence of detected discrepancy, with the reported low power making a stronger interpretation especially unwarranted.

Appendix B. Additional ETF experiment: implementation and provenance

A complementary exploratory experiment evaluated eight fixed rule sets on DIA, MDY, EWG, EWJ, EWU, EWC, EWA, and EWH over 2008-2025: RSI(2), Donchian20/10, IBS, Bollinger20/2, MACD12/26/9, SMA50/200, monthly 252-session momentum, and monthly top-three 12–1 momentum rotation. Signals used closing information, with execution at the next opening and explicit transaction costs. Net trade outcomes were scaled by a two-ATR risk unit fixed before entry; exits remained strategy-specific. RSI(2) produced 1,023 trades and crossed Gate 1 marginally ($t = 3.054$), but exceeded the instrument reference threshold of Gate 2; its net-return t-statistic was 2.948. The other seven rules did not reach $t = 3$ on R. Instrument groups were insufficient for full Gate 2 assessment in SMA, monthly momentum, and rotation. Gate 2 used exact empirical Wasserstein distance, 10,000 Gaussian simulations, a previous-session VIX boundary of 20, and 2008-2013, 2014-2019, and 2020-2025 time blocks. No ETF rule cleared the combined rule; for the three slower rules, Gate 2 was incomplete because P1 was insufficient. These post-hoc results remain exploratory: Gaussian calibration is not distribution-free, trade-level t-statistics ignore dependence, and non-rejection does not establish invariance.

The eight-rule panel was assembled after earlier experiments had been inspected; it is not an untouched out-of-sample validation. Trading costs are 0.005 USD per share with a one-dollar minimum per order and one basis point of slippage per side. Each position is sized using a 0.25 percent capital risk budget and a 10 percent notional cap; the two-ATR unit is not a guaranteed stop. All rules are long-only. Monthly strategies retain eligible positions rather than generating artificial monthly round trips. RSI and Donchian use their separately recorded experiment; the six-rule panel uses the v2 previous-session VIX availability check. Complete execution and coverage flags were reported for all eight runs. Trade-level CSVs have been recovered for IBS (2,706 trades), Bollinger (756), and MACD (1,466). Each file was checked for duplicate trade identities, date order, finite numerical values, positive risk units, nonnegative fees, and reconciliation of R with net PnL divided by the risk unit. Recomputing the mean, trade-level t-statistics, all three SIS components, and the 10,000-draw Gaussian reference thresholds reproduced the earlier reports within a numerical tolerance of 10^{-12}. This is a consistency check of the supplied outputs and test calculations; it is not an independent replay of source market data or simulated fills. RSI and Donchian trade exports remain to be recovered. The three slower-rule results are retained as reported, with P1 marked insufficient; their trade exports have not been collected.

Earlier diversified-ETF RSI experiment. A separately recorded exploratory RSI(2) run used SPY, QQQ, IWM, EFA, EEM, TLT, GLD and XLF. It reported $N = 1{,}035$, mean $R = 0.09623$ and $t(R) = 4.290$, clearing Gate 1. Its instrument component was below its reference (0.177291 versus 0.194566), while the regime component (0.160317 versus 0.156116) and time component (0.152414 versus 0.140973) exceeded theirs. This run used 2,000 Gaussian simulations. Its different universe and simulation budget distinguish it from the eight-rule index-ETF panel, so its values are not pooled into Table 3. The trade-level CSV has not been recovered; the result is retained as a recorded exploratory output.

A separate option-straddle attempt produced 452 closed trades from 504 planned observations, with 52 data issues and status DATA_INCOMPLETE. It is not included in the complete ETF comparison and is not used to support a performance or invariance claim.

Table 5 ETF distributional scores and recorded reference thresholds

	P1: instrument		P2: regime		P3: time	
Rule	SIS	$\widehat{\tau}_{P1}$	SIS	$\widehat{\tau}_{P2}$	SIS	$\widehat{\tau}_{P3}$
RSI(2)	0.200093	0.194403	0.092498	0.161714	0.127177	0.141809
Donchian20/10	0.167965	0.219216	0.246033	0.179393	0.113027	0.160285
IBS pullback	0.100188	0.120905	0.116169	0.104062	0.095812	0.087737
Bollinger20/2	0.128731	0.224235	0.071785	0.157567	0.205381	0.163219
MACD12/26/9	0.134498	0.162490	0.265450	0.116340	0.151227	0.118754
SMA50/200	–	–	0.489592	0.409379	0.645379	0.434989
Momentum252 monthly	–	–	0.413464	0.425366	0.397126	0.432707
Rotation12–1 top3	–	–	0.397254	0.391438	0.285405	0.378159

Notes: Values are copied from the recorded ETF experiment reports and rounded to six decimal places; no statistic or threshold is recalculated. The reference thresholds are the recorded 95th percentiles from 10,000 Gaussian simulations. A component clears its recorded reference when SIS is strictly below its threshold. Dashes denote insufficient instrument-group data, not a distance of zero. Available P2 and P3 components are shown even when P1 is insufficient, without assigning a complete Gate 2 pass. RSI and Donchian are recorded notebook summaries; IBS, Bollinger and MACD are additionally supported by the recovered trade files and their numerical verification reports.

Appendix C. Mathematical properties of the score

The following statements concern the exact empirical score in Equation (7). They make explicit standard consequences of Wasserstein geometry and elementary probability. They are not finite-sample guarantees for the historical interpolated FX approximation or for dependent market observations.

Proposition 1 (Common-affine invariance and the zero case) *Fix one partition with $K \geq 2$ nonempty groups and a finite outcome vector r with sample standard deviation $s_r > 0$. Write*

$$S(r) = \frac{1}{\binom{K}{2} s_r} \sum_{j<k} W_1(\widehat{F}_j, \widehat{F}_k).$$

Then $S(r) \geq 0$, and $S(r) = 0$ if and only if all group empirical probability measures are identical. For every common $a \neq 0$ and $b \in \mathbb{R}$,

$$S(ar + b) = S(r).$$

Proof. Every summand is nonnegative. Since W_1 is a metric on probability measures with finite first moments, their sum is zero exactly when every pair of empirical laws agrees. Under $T(x) = ax + b$, any coupling of two laws gives a coupling of their images with transport cost multiplied by $|a|$. Applying the inverse transformation gives the reverse inequality, hence $W_1(T_{\#}\mu, T_{\#}\nu) = |a| W_1(\mu, \nu)$. Here $T_{\#}\mu$ denotes the distribution of $T(U)$ when $U \sim \mu$. The pooled standard deviation is also multiplied by $|a|$, proving the claim. This argument uses a single common transformation, not observation-specific risk denominators. □

Proposition 2 (Payoff differences controlled by the distance) *For empirical group laws $\widehat{F}_j, \widehat{F}_k$ and any L-Lipschitz function $h : \mathbb{R} \to \mathbb{R}$,*

$$\left| \int h \, d\widehat{F}_j - \int h \, d\widehat{F}_k \right| \leq L W_1(\widehat{F}_j, \widehat{F}_k). \tag{12}$$

Consequently,

$$S(r) \geq \frac{1}{\binom{K}{2} s_r} \sum_{j<k} |\bar{r}_j - \bar{r}_k|.$$

Proof. For any coupling (U, V) of the two empirical laws,

$$|\mathbb{E}h(U) - \mathbb{E}h(V)| \leq \mathbb{E}|h(U) - h(V)| \leq L\mathbb{E}|U - V|.$$

Taking the infimum over couplings gives Equation (12). Set $h(r) = r$, for which $L = 1$, and average over pairs to obtain the mean bound. Taking $h(r) = (-r)_+ = \max\{-r, 0\}$, also 1-Lipschitz, gives the analogous bound for expected downside magnitude. A finite-sample score below a reference threshold still does not provide an upper confidence bound for these population quantities. □

Theorem 1 (Population limit under an idealised sampling model) *For one fixed partition, suppose* $(R_i, L_i)_{i\geq 1}$ *are independent and identically distributed,* $L_i \in \{1, \ldots, K\}$, $p_j = \Pr(L_i = j) > 0$ *for every group, and* $\mathbb{E}[R_i^2] < \infty$, *with* $\sigma_R^2 = \mathrm{Var}(R_i) > 0$. *Let* $F_j = \mathcal{L}(R_i \mid L_i = j)$. *Then, almost surely,*

$$S_N \longrightarrow D = \frac{1}{\binom{K}{2}\sigma_R} \sum_{j<k} W_1(F_j, F_k). \tag{13}$$

If at least two conditional laws differ, $D > 0$. *An ideal Gaussian reference quantile with fixed tail probability converges to zero as all group sizes increase, so the probability that the score exceeds this quantile tends to one under that fixed alternative. The same detection statement holds for a fixed finite number B of Gaussian reference simulations and their empirical quantile.*

Proof. The strong law gives $n_j/N \to p_j > 0$, so every group size diverges. Each conditional empirical law converges almost surely to F_j in W_1: weak empirical convergence and convergence of the first absolute moment give Wasserstein convergence. The triangle inequality therefore yields

$$|W_1(\widehat{F}_j, \widehat{F}_k) - W_1(F_j, F_k)| \leq W_1(\widehat{F}_j, F_j) + W_1(\widehat{F}_k, F_k) \to 0.$$

The pooled sample standard deviation converges to σ_R, proving Equation (13). Condition on a label sequence whose group shares have positive limits, as occurs almost surely under the stated model. Under the Gaussian reference, every group then contains a growing standard-normal sample. All group laws have the same limit, and the reference score converges to zero in probability. Its fixed $(1-\alpha)$-quantile consequently converges to zero for $0 < \alpha < 1$. For fixed B, the maximum of B reference scores also converges to zero in probability by a finite union bound; their interpolated empirical quantile is bounded by this maximum. Since the observed score tends to $D > 0$, exceedance probability tends to one. □

This theorem describes fixed alternatives with growing groups. It does not provide a convergence rate, a universal N cutoff, finite-sample power, or Gaussian-reference size control under a non-Gaussian equality null. Independence of the pairs is an idealisation; deterministic historical time partitions and overlapping trade outcomes require a sampling model appropriate to their construction.

Proposition 3 (Conditional Gaussian location-scale reference) *Fix the observed labels independently of the outcome randomness. Suppose, conditionally on those labels, the outcomes are independent* $\mathcal{N}(\mu, \sigma^2)$, *with* $\sigma > 0$. *Then the joint distribution of the exact score vector* $(\mathrm{SIS}_{d,N})_{d\in\mathcal{D}}$ *does not depend on* μ *or* σ, *and can be simulated using independent standard-normal outcomes with the same labels.*

Proof. Write $R_i = \mu + \sigma Z_i$, where the Z_i are independent standard-normal variables. Proposition 1 removes the same common location and scale from every dimension simultaneously. Thus the resulting vector of scores has the same law as the one computed from (Z_i) with the fixed labels. □

This statement justifies standard-normal simulation within the Gaussian location-scale family. It does not replace an arbitrary common outcome distribution by a Gaussian law. The reported Monte Carlo quantiles are reference thresholds under this model. No alternative calibration, family-wise adjustment or equivalence decision rule is applied in this revision.